

No. of Printed Pages : 4
Roll No.

188543

**Level -4, 2nd Sem/ DVOC
(Medical Imaging Technology, SD)
Subject : Digital Electronics**

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple choice questions. All questions are compulsory (5x1=5)

- Q.1 Digital Signal is _____ in Nature
a) Continuous b) Discrete
c) Both 'a' & 'b' d) None
- Q.2 $A.1 =$ _____
a) 0 b) 1
c) A d) None
- Q.3 In NAND gate AND is followed by
a) NOR b) OR
c) NOT d) Ex-OR
- Q.4 Half adder has _____ number of inputs.
a) 1 b) 2
c) 3 d) 4
- Q.5 7-Segment display, displays _____
a) 0-10 b) 0-7
c) 0-9 d) 1-9

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

- Q.6 What do you mean by analog signal?
- Q.7 One byte = _____ bits
- Q.8 How many select lines are used in 1:16 MUX?
- Q.9 Expand MSI.
- Q.10 Define Race Around Condition.

SECTION-C

Note: Very Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

- Q.11 State and explain De-Morgan's law.
- Q.12 Explain 7-Segment display in details.
- Q.13 What are the differences between static and dynamic memory?
- Q.14 Write a short note on VLSI.
- Q.15 Explain the applications of Linear ICs.
- Q.16 Explain half adder in detail with truth table.
- Q.17 What is flip flop? Explain D-Flip flop in details.
- Q.18 What are the differences between analog and digital signal.

SECTION-D

Note: Long answer type questions. Attempt any one question out of two questions. (1x10=10)

- Q.19 Explain a 4-bit serial input shift register in details and give its timing diagram.
- Q.20 What is universal gate? Design AND, OR and NOT gate using NOR gate with their truth table.